La Scuola d'Italia / Huson / Physics Nickname:
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2.5 Classwork: Falling Objects

Reference (provided). For a falling object, the acceleration is $g = 9.80$ m/s^2, downward.

$$v = v_0 + at \qquad v^2 = v_0^2 + 2a(y - y_0)$$

$$y = y_0 + v_0 t + \tfrac{1}{2}at^2 \qquad \bar{v} = \frac{v + v_0}{2}$$

Ignore air resistance unless the problem says otherwise. "Dropped" means $v_0 = 0$. Choose which direction is positive, up or down, and keep it for the whole problem. List what you know and what you want, then choose the equation.

1. A ball is dropped from a tall building. Take *down* as positive, so $a = +9.80$ m/s^2, and $y_0 = 0$.

   (a) Fill in the table.

t (s)	0	1.00	2.00	3.00
v (m/s)				
y (m)				

   (b) At 2.00 s, the ball has fallen how many times as far as at 1.00 s? At 3.00 s? What pattern do you see?

2. Galileo is said to have dropped balls from the Leaning Tower of Pisa, which is 55.86 m tall on its low side. Take down as positive.

   (a) How long does a ball dropped from the top take to reach the ground? (Solve for t^2 first, then take the square root.)

   (b) How fast is it going when it lands?

   (c) A heavy iron ball and a light wooden ball of the same size are dropped together. Which lands first?

3. A student throws a ball straight up. It leaves the hand at 14.7 m/s. Take *up* as positive, so $v_0 = +14.7$ m/s and $a = -9.80$ m/s^2.

   (a) At the highest point, what is the ball's velocity? What is its acceleration?

   (b) How long does the ball take to reach the highest point?

   (c) How high above the hand does it go?

   (d) How long after the throw does the ball come back to the hand, and what is its velocity then?

4. A raindrop falls from a cloud 1000 m above the ground. Take down as positive.

   (a) If there were no air, how fast would the raindrop be going when it reached the ground?

   (b) Real raindrops land at about 9 m/s. Why is that so much slower?